EXAM SOLUTIONS

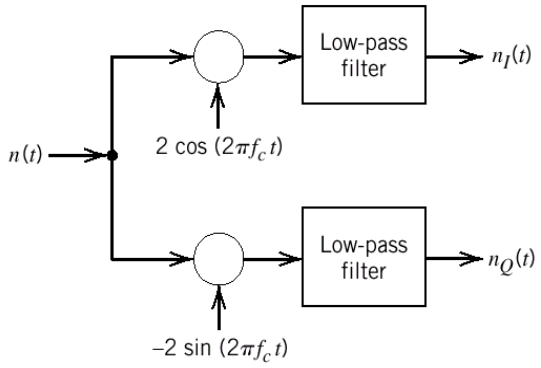
1. a) **This question is straightforward bookwork, and most students have done well.**

i) [2]

A coherent receiver is phase-locked to the transmitter, while in the case of a non-coherent receiver there is no synchronization between the transmitter and the receiver.

ii) [3]

$$n(t) = n_I(t)\cos(2\pi f_c t) - n_Q(t)\sin(2\pi f_c t).$$



iii) [2]

The SNR performance of DSB-SC is the same as that of baseband transmission.

iv) [3]

The SNR performance of FM is significantly higher than DSB-SC, but it also requires a higher transmission bandwidth.

b) **This question requires an understanding of the properties of the autocorrelation function of WSS processes, which were stated in the class, and repeated in the revision lecture. Based on the scores, this was by far the hardest question in the exam (answered by the least number of students). It seems that most students wanted to check whether the autocorrelation function is a function of the time difference or not, and did not think about the other properties, i.e., even and takes its maximum value at 0.**

i) [2]

No. $R_X(t_1,t_2)$ should depend only on $t_1 - t_2$ for a WSS process.

ii) [3]

No. We should have $R_X(t_1,t_2) = R_X(t_2,t_1)$.

iii) [4]

No. We should have $R_X(\tau) \leq R_X(0)$ for all τ.

c) **This question requires the application of basic concepts on sampling and quantization. While many students answered most parts of the question correctly, the typical mistake was to try to use the $\Delta^2/12$ formula for the high rate quantization noise variance. Note that the problem constrains the maximum quantization noise (not its variance), which directly leads to a bound on the number of quantization intervals.**

i) [2]

From the Nyquist theorem the minimum sampling rate is 8 KHz.

ii) [3]

Maximum quantization error for a uniform quantizer is equal to $\Delta/2$, i.e., half the size of the quantization region. Then we have

$$\frac{\Delta/2}{L \cdot \Delta} \leq 0.004,$$

where L is the number of quantization regions. We get $L \geq 125$. We need a minimum of 7 bits per source sample.

iii) [2]

Bit rate is $7 * 8 = 56$ KHz.

iv) [3]

We should satisfy

$$56K \text{ bits/sec} = \frac{4 \text{ bits/symbol}}{T \text{ sec/symbol}} = \frac{4}{T} \text{ bits/sec}$$

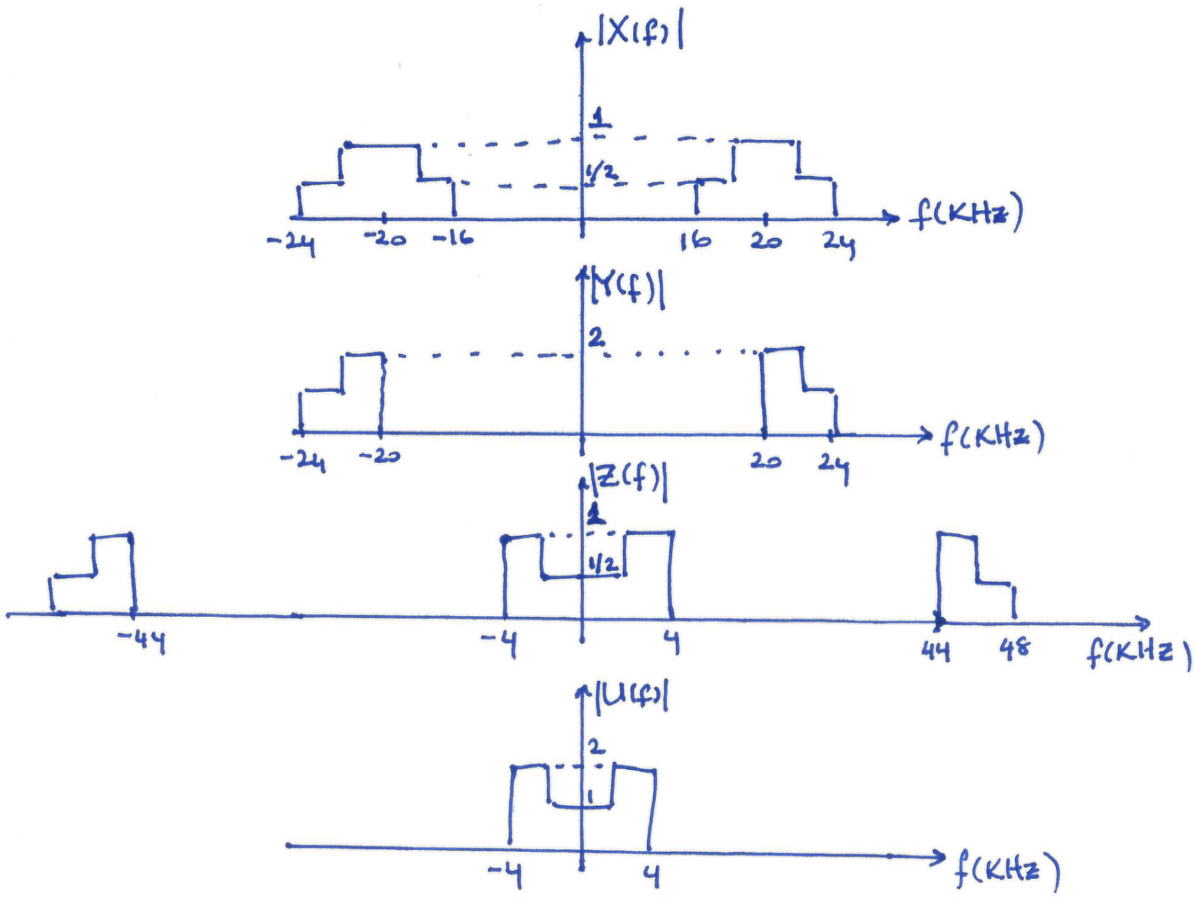
From here we get $T \approx 71.4$ microseconds.

d) **This question requires a basic understanding of modulation, and its effect on the spectrum, and does not involve any noise analysis. While most students answered this question at least partially correctly, the typical mistake was applying the filters.**

Also note that, once the first part is answered, it is easy to see the impact of the system on the spectrum of the input signal. With this observation the second part can be answered directly without going through the steps once again.

i) [5]

The magnitude spectrum of signals $x(t)$, $y(t)$, $z(t)$ and $u(t)$ are as follows:



ii) [3]

The system simply reversed the spectrum of $m(t)$. If we apply the same on $u(t)$, we obtain $m(t)$.

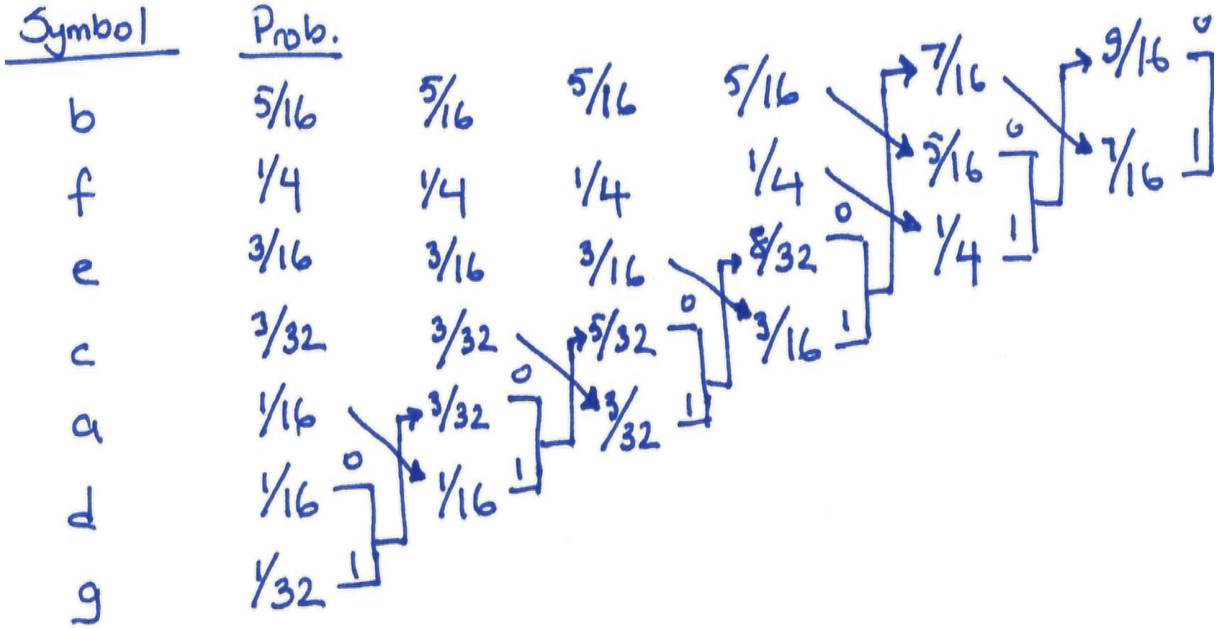
iii) [2]

$$w(t) = m(t).$$

2. a) **This question requires understanding the algorithm that generates the Huffman codewords. The students should also know how to compute the entropy of a memoryless source, and its relation to the average codeword length of any given source code.**

i) [8]

We order the symbols from the most likely to the least likely, and apply the Huffman algorithm as below.



Then we obtain the following codewords for each symbol:

Symbol	Probability	Codeword
a	1/16	1001
b	5/16	00
c	3/32	101
d	1/16	10000
e	3/16	11
f	1/4	01
g	1/32	10001

ii) [2]

The average codeword length is given by

$$4\times\frac{1}{16}+2\times\frac{5}{16}+3\times\frac{3}{32}+5\times\frac{1}{16}+2\times\frac{3}{16}+2\times\frac{1}{4}+5\times\frac{1}{32}=2.5 \text{ bits per sample.}$$

iii) [4]

The entropy of the source is found as

$$\begin{aligned} H(S) &= \frac{1}{16}\log_2 16 + \frac{5}{16}\log_2 \frac{16}{5} + \frac{3}{32}\log_2 \frac{32}{3} + \frac{1}{16}\log_2 16 \\ &\quad + \frac{3}{16}\log_2 \frac{16}{3} + \frac{1}{4}\log_2 4 + \frac{1}{32}\log_2 32 \\ &= 2.45 \text{ bits per sample.} \end{aligned}$$

iv) [2]

The code is not optimal since its rate is slightly below the entropy bound.

b) **Similar problems have been solved in the class. It is slightly harder than the ones done in the class, which typically consisted of only one sinusoidal. The key is to be able to write down the autocorrelation function, and then to use the independence of the two random variables. Some students, trying the repeat the examples in the lectures, tried to take expectation over f_c although it is a constant in this problem. Another common mistake was in the manipulation of the trigonometric identities. Although these are provided to the students as part of the exam paper, many have made mistakes by not clearly writing down the expressions, confusing a plus with a minus, etc.**

Most students have correctly stated the requirements for a random process to be WSS, for which they were given partial credits even if the first part of their answer was not correct.

i) [5]

We have

$$\begin{aligned} E[X(t)] &= E[N_1\cos(2\pi f_c t) - N_2\sin(2\pi f_c t)] \\ &= E[N_1]\cos(2\pi f_c t) - E[N_2]\sin(2\pi f_c t) \\ &= 0 \end{aligned}$$

$$\begin{aligned} R_X(t_1,t_2) &= E[X(t_1)X(t_2)] \\ &= E\left[(N_1\cos(2\pi f_c t_1) - N_2\sin(2\pi f_c t_1))(N_1\cos(2\pi f_c t_2) - N_2\sin(2\pi f_c t_2))\right] \\ &= E[N_1^2]\cos(2\pi f_c t_1)\cos(2\pi f_c t_2) + E[N_2^2]\sin(2\pi f_c t_1)\sin(2\pi f_c t_2) \\ &\quad - E[N_1N_2](\cos(2\pi f_c t_1)\sin(2\pi f_c t_2) + \sin(2\pi f_c t_1)\cos(2\pi f_c t_2)) \\ &= \sigma^2[\cos(2\pi f_c t_1)\cos(2\pi f_c t_2) + \sin(2\pi f_c t_1)\sin(2\pi f_c t_2)] \\ &= \sigma^2\cos(2\pi f_c(t_1 - t_2)), \end{aligned}$$

where we have used the linearity of the expectation operation, and the independence of N_1 and N_2, i.e., $E[N_1N_2] = E[N_1]E[N_2] = 0$. We can write the autocorrelation function as follows:

$$R_X(\tau) = \sigma^2\cos(2\pi f_c\tau).$$

ii) [2]

Since the mean is independent of time, and the autocorrelation function depends only on the time difference, $X(t)$ is WSS.

iii) [3]

We have

$$\begin{aligned} S_X(f) &= \mathbb{F}(R_X(\tau)) \\ &= \mathbb{F}\left(\sigma^2 \cos(2\pi f_c \tau)\right) \\ &= \frac{\sigma^2}{2}[\delta(f-f_c)+\delta(f+f_c)]. \end{aligned}$$

iv) [4]

We can consider $Y(t)$ as the output of an LTI filter with frequency response $H(f) = (j2\pi f)^n$. Then the PSD of $Y(t)$ is found as:

$$\begin{aligned} S_Y(f) &= |H(f)|^2 S_X(f) \\ &= |j2\pi f|^{2n} S_X(f) \\ &= \frac{\sigma^2}{2}(2\pi f_c)^{2n}[\delta(f-f_c)+\delta(f+f_c)]. \end{aligned}$$

3. a) **This problem requires a good understanding of the probability of error calculation in digital communication systems, and the application of the Leibnitz rule for differentiation, which was provided. Some students tried to use directly the formula obtained in the class for the error probability, without noticing that it was for a unipolar system, while the exam question considers a bipolar system.**

i) [2]

Due to the symmetry in the system, the optimal threshold in this case is trivial, and is given by 0.

ii) [6]

Probability of error when a "0" is transmitted is given by

$$P_{e0} = \int_{1+T}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2},$$

where T is the threshold of the detector.

Probability of error when a "1" is transmitted is given by

$$P_{e1} = \int_{1-T}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2}.$$

The overall probability of error can be written as

$$P_e = p_0 \int_{1+T}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2} + p_1 \int_{1-T}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2}.$$

The optimal threshold is found by solving for $\frac{dP_e}{dT} = 0$. We use the Leibnitz rule to take the derivative, and obtain the following:

$$\frac{dP_e}{dT} = -p_0 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(1+T)^2/2\sigma^2} + p_1 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(1-T)^2/2\sigma^2} = 0.$$

From here we obtain

$$\ln \frac{p_0}{p_1} = \frac{(1+T)^2 - (1-T)^2}{2\sigma^2} = \frac{2T}{\sigma^2} = 4T.$$

We find $T = \frac{\ln 4}{4} \approx 0.35$.

iii) [6]

The error probability can be written as

$$P_e = p_0 \cdot Q\left(\frac{1+T}{\sigma}\right) + p_1 \cdot Q\left(\frac{1-T}{\sigma}\right).$$

The receiver uses the T from above, given by $T = 0.35$. Then we have

$$\begin{aligned} P_e &= 0.3 \cdot Q(2.7) + 0.7 \cdot Q(1.3) \\ &\approx 0.3 \times 2 \times 10^{-3} + 0.7 \times 1 \times 10^{-1} \\ &\approx 0.07006. \end{aligned}$$

This is more than three times the probability of error the receiver was expecting to achieve, which is found to be

$$\begin{aligned} P_e &= 0.8 \cdot Q(2.7) + 0.2 \cdot Q(1.3) \\ &\approx 0.8 \times 2 \times 10^{-3} + 0.2 \times 1 \times 10^{-1} \\ &\approx 0.0216. \end{aligned}$$

b) **This problem requires a good understanding of the Fourier transform, and the impact of modulation on the spectrum. Students also need to know/ derive the modulator/demodulator characteristics of an SSB system, and be able to analyse its SNR behaviour in the presence of noise.**

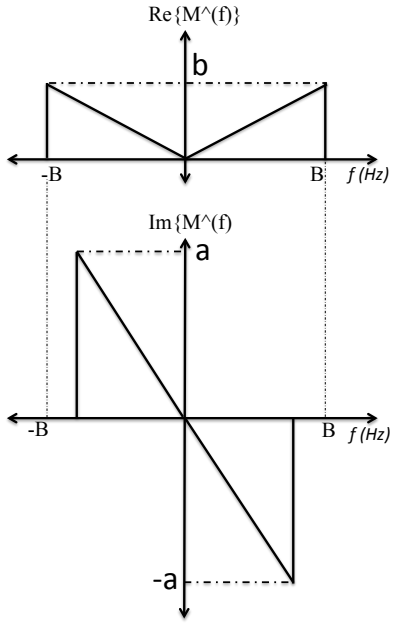
While many students were able to answer part i) and obtained $\hat{M}(f)$, mistakes in plotting its real and imaginary components were common. This requires carefully evaluating the impact of Hilbert transform on these two components (mainly manipulation of complex numbers). Part iii) is very similar to Question 1-d, while the last two parts were direct application of course knowledge, and were answered by the majority.

i) [2]

$\hat{m}(t) = m(t) * \frac{1}{\pi t}$.

ii) [4]

$\hat{M}(f) = -j \cdot sgn(f) \cdot M(f)$



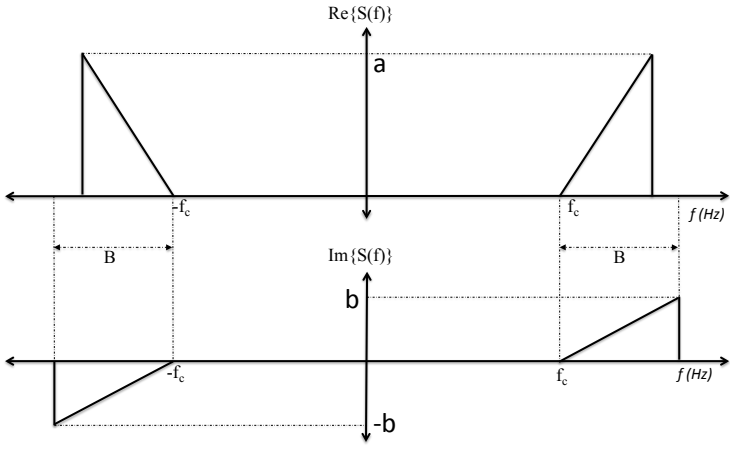
iii) [4]

The students may remember from class that the USB-SSB transmits only the upper halves of the spectrum of the signal, centered around carrier frequency f_c.

They should also be able to derive this directly from the basic principles as follows:

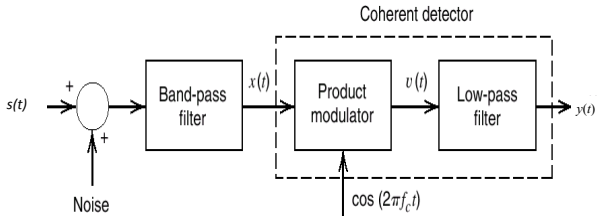
$$S(f) = \frac{1}{2}\left[M(f-f_c)+M(f+f_c)\right] - \frac{1}{2j}\left[\hat{M}(f-f_c)-\hat{M}(f+f_c)\right]$$

By shifting the spectrum of $M(f)$ and $\hat{M}(f)$, and applying the appropriate sign and conjugate adjustments, we get the spectrum below:



iv) [4]

The coherent receiver diagram is as follows:



At the output of the bandpass filter we get $x(t) = s(t) + n(t)$, where $n(t)$ is the bandpass noise. We have

$$x(t) = m(t)\cos(2\pi f_c t) - \hat{m}(t)\sin(2\pi f_c t) + n_I(t)\cos(2\pi f_c t) - n_Q(t)\sin(2\pi f_c t).$$

At the end of the demodulator we obtain

$$y(t) = m(t) + n_I(t).$$

v) [2]

SSB-SC has the same SNR performance as DSB-SC; however, it uses only half the bandwidth of DSB-SC.